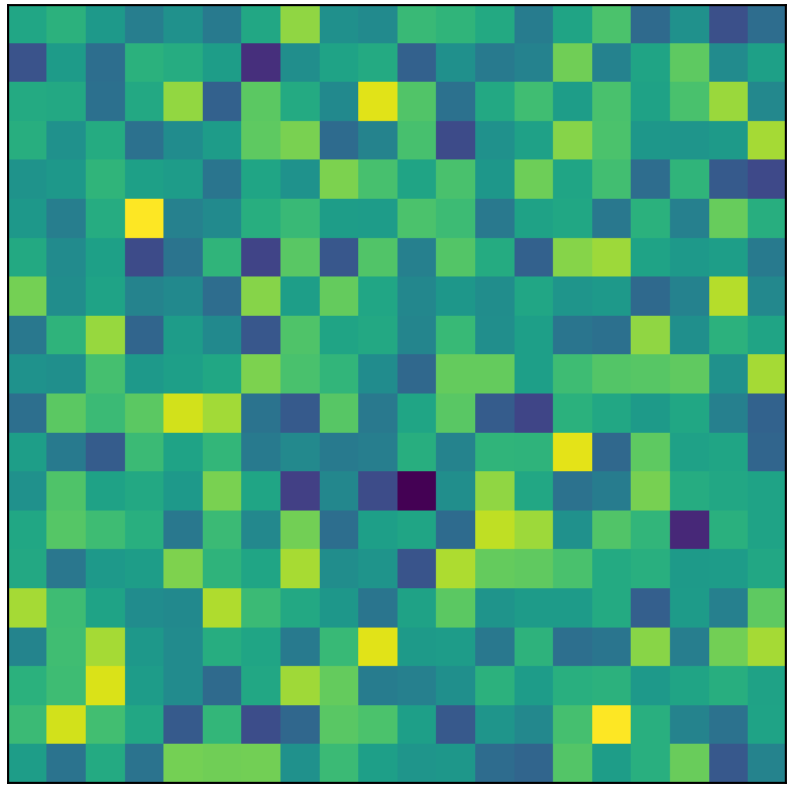
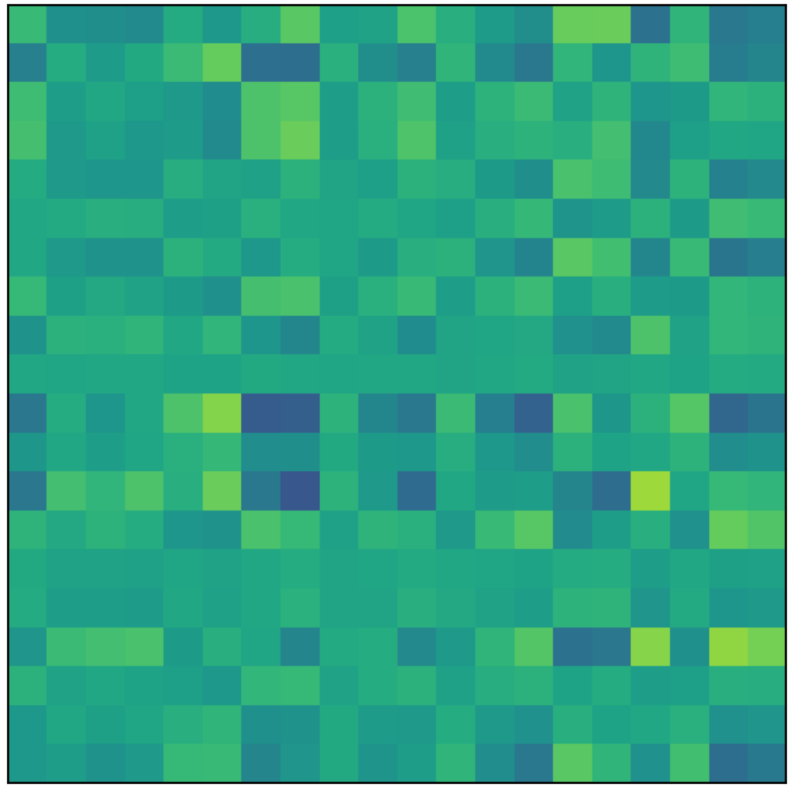
LoRA rank sweep on random full-rank shift
$M = 20$, $D = M^2 = 400$, benchmark=sphere
matched search dims: r=1: d=40, r=2: d=80, r=4: d=160

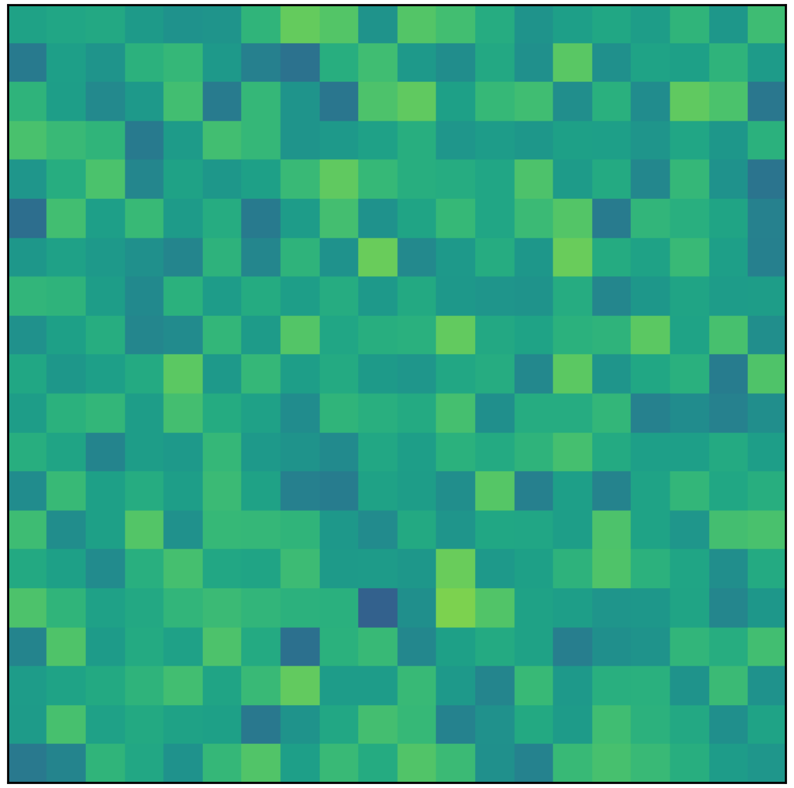
Random shifted optimum X_*



LoRA SVD at r=2, $d = 2Mr = 80$ ($A \in \mathbb{R}^{20 \times 2}$, $B \in \mathbb{R}^{2 \times 20}$)
rel. MSE=7.06e-01



Best RP at r=2, $d = 80$ ($P \in \mathbb{R}^{80 \times 400}$)
rel. MSE=7.25e-01



Setup

Full space: $D = M^2 = 400$.

At each LoRA rank r, all methods search in the same dimension $d = 2Mr$:

```
LoRA: rank-r factors (A,B), d=2Mr
RP:   random projection P, d=2Mr
RB:   random blocking groups, d=2Mr

Per-rank search dimensions:
  r=1  ->  d=40
  r=2  ->  d=80
  r=4  ->  d=160
```

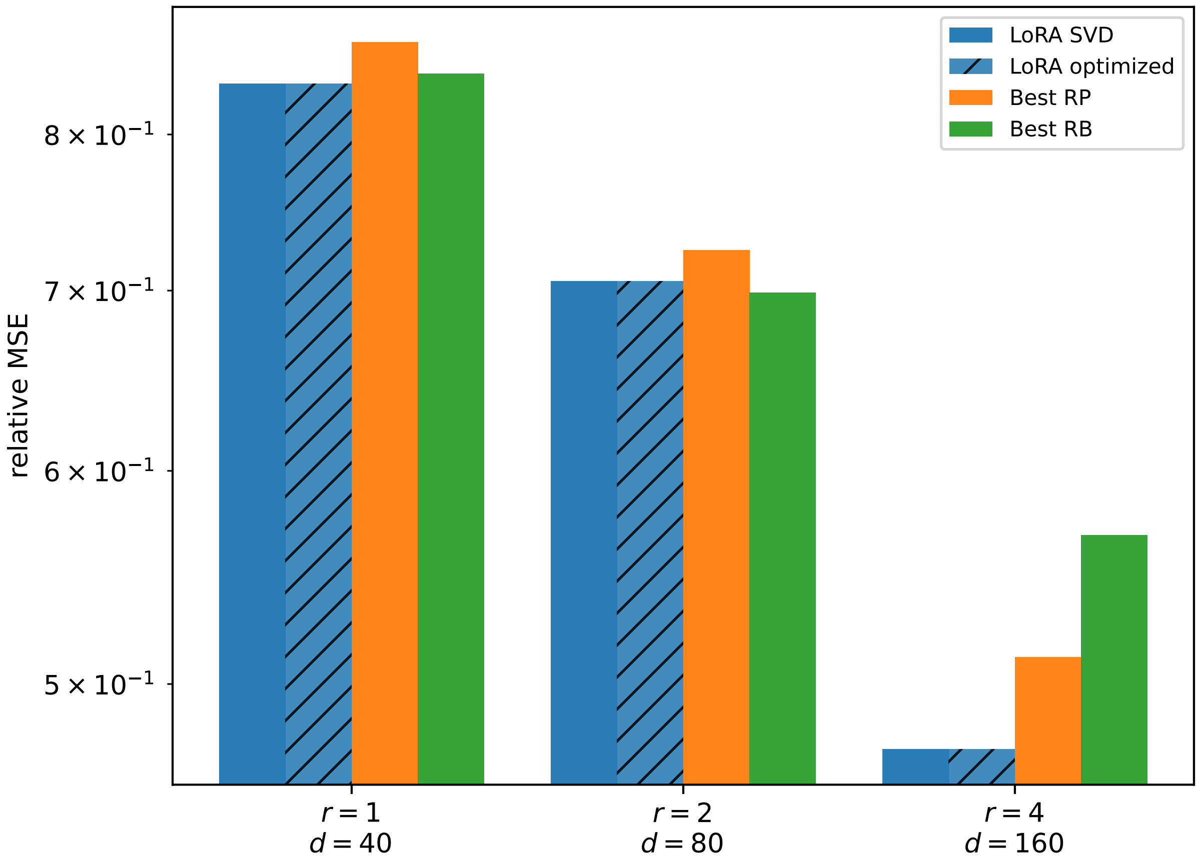
Random full-rank optimum.

Higher LoRA rank improves the best rank-r approximation (SVD lower bound).

RP/RB baselines are recomputed at each matched search dimension.

At r=2, RB uses $d = 80$ ($D = 400 \rightarrow \{0, \ldots, 79\}$).

Reconstruction error vs rank (matched d per method)



Shifted sphere vs rank (matched d per method)